

Designing and Running Experiments: From the Laboratory to the Field

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Notes: overview rather than a comprehensive account of the field

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- I shed light on applied microeconomics questions via experiments
 - Behavioral Economics $\cap$ Experimental Economics

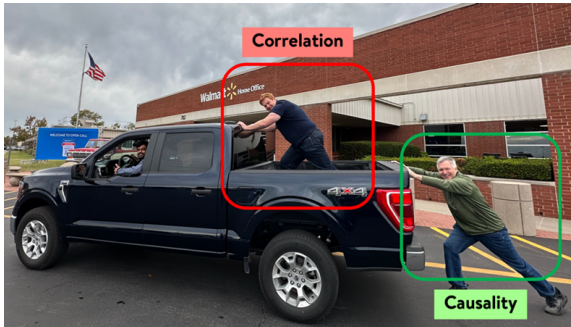
- ① Ode to Causality
- ② Anatomy of an Experiment
- ③ Failures and applied tips in the field
- ④ Future directions & Hot topics

Ode to Causality

Correlation vs. Causation

Correlation: A statistical relationship between two variables where changes in one variable are associated with changes in another.

Causality: refers to a direct cause-and-effect relationship where changes in one variable lead to changes in another.



Correlation does not imply causation!

Obsession for Causality

- Need to go beyond correlational analyses in social sciences
- Economists have long used precise models and econometric techniques to answer causal questions using variations in naturally occurring data

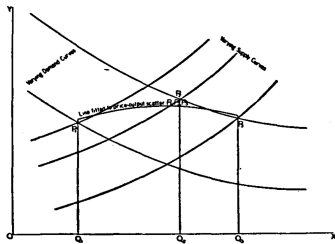


Figure: *Exogenous shocks, Ceteris-Paribus*

- More recently: use of EXPERIMENTS!

Trends in Econ Publications

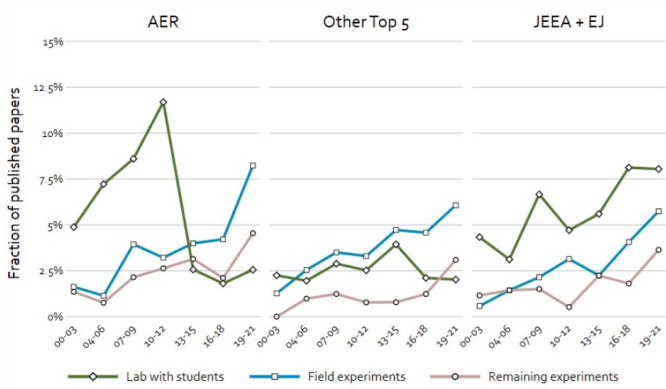


Figure A1 – Fraction published experimental articles per year

Figure: from Reubens et al. 2022

Experiments 101

Treatment Effects and Randomization

Treatment Effect & Identification

Objective: evaluate the effect of an intervention (e.g., policy)

- Y_i is our outcome of interest measured on person i
- Potential Outcome: if i receives the treatment we observe $Y_i(1)$, otherwise $Y_i(0)$
- We measure our *Treatment effect* using the **mean difference estimator**:

$$\tau_i = Y_i(1) - Y_i(0)$$

- Average Treatment Effect (ATE):

$$E[Y_i(1)] - E[Y_i(0)] = \frac{1}{N} \sum_i \tau_i$$

Example (ideal world)

Chattopadhyay & Duflo 2004: Village Council head positions in India have been randomly reserved for a woman. Authors study the public expenditure when having women, rather than men, as head Indian village councils

<i>Village i</i>	$Y_i(0)$ <i>Budget share if village head is male</i>	$Y_i(1)$ <i>Budget share if village head is female</i>	τ_i <i>Treatment effect</i>
Village 1	10	15	5
Village 2	15	15	0
Village 3	20	30	10
Village 4	20	15	-5
Village 5	10	20	10
Village 6	15	15	0
Village 7	15	30	15
Average	15	20	5

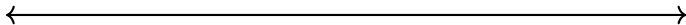
$$E[Y_i(1)] - E[Y_i(0)] = 20 - 15 = 5$$

Counterfactual

- The fundamental difficulty that arises is that **we do not observe both states of the world**
- We need a **counterfactual** as benchmark
- **Key question:** how to construct a proper counterfactual?
 - **Randomization**, (controlled experiments, e.g., laboratory, field, natural experiments)
 - Quasi-Randomization (e.g., quasi-experiment)
 - Others: Propensity score matching, Instrumental variables

Controlled Data

Naturally Occurring Data



Lab

Field

Quasi-E

PSM, IV

An (realistic) Example

Randomization as solution to the counterfactual issue

<i>Village i</i>	$Y_i(0)$ <i>Budget share if village head is male</i>	$Y_i(1)$ <i>Budget share if village head is female</i>	τ_i <i>Treatment effect</i>
Village 1	?	15	?
Village 2	15	?	?
Village 3	20	?	?
Village 4	20	?	?
Village 5	10	?	?
Village 6	15	?	?
Village 7	?	30	?
Estimated average based on observed data	16	22.5	6.5

Randomization

If the treatment is administered *randomly* (all villages have equal chances to be treated) then, the treatment group is a random sample of all villages, and therefore they all **share the same characteristics**, but **only differ** in the presence of the intervention administered.

ATE : Some boring maths

Proposition 1: Under randomization, potential outcomes are equivalent

Proof: Formally, let us indicate the treatment group with $T = 1$, $T = 0$ otherwise:

$$E[Y_i(1)|T = 1] = E[Y_i(1)] \quad (1)$$

$$E[Y_i(1)|T = 0] = E[Y_i(1)] \quad (2)$$

hence

$$E[Y_i(1)|T = 1] = E[Y_i(1)|T = 0] \quad (3)$$

similarly

$$E[Y_i(0)|T = 0] = E[Y_i(0)|T = 1] = E[Y_i(0)] \quad (4)$$

We can rewrite the ATE as:

$$E[Y_i(1)|T = 1] - E[Y_i(0)|T = 0] \quad (5)$$

What if no randomization?

Proposition 2: Under no randomization, we incur into biased estimates

Proof: Assume that the treatment is given not at random, but it depends, for example, on some village characteristics (selection bias)

ATE becomes (by adding and subtracting $E[Y_i(0)|T = 1]$) :

$$E[Y_i(1)|T = 1] - E[Y_i(0)|T = 1] + E[Y_i(0)|T = 1] - E[Y_i(0)|T = 0] \quad (6)$$

- the red component is the ATE among the treated (ATT)
- the blue component is the selection bias
- under randomization the selection bias is zero
- when treatment is not given at random, the bias is different than zero

Randomization procedure: an example

- All units have equal chances to end up being treated
- *Simple random assignment*: Flip a coin; problem: small chance to end up with $N/2$ obs. in the treatment
- *Complete random assignment*: set m of N subjects to be included in the treatment with equal probability
 - Flip a coin until you get m units in the treatment
 - Enumerate all subjects, then compute all possible ways to group m subjects, and randomly select one
 - Rely on a statistical software

```
1 set.seed(1234567) # set random seed
2 N <- 7             # number of units
3 m <- 2             # number of treated units
4 sel <- rnorm(N)     # N standard Normal deviates
5 Tr <- order(sel)    # ordering permutation which orders the deviates
6 Tr[1:m]            # the first m indexes are treated units
7
```

Anatomy of an Experiment

Taxonomy (Harrison & List 2004)

- *Conventional Lab Experiment*: student pools, abstract framing, imposed set of rules
- *Artefactual field experiment*: same as Conventional Lab but with non-student pools
- *Framed field experiment*: same as Artefactual but with field context in either the task or info subjects can use
- *Natural field experiment*: same as Framed but environment is where subjects naturally undertake tasks, and they do not know to be in an experiment (naturally occurring experiments)

Note: some experiments may share features of more categories

A Taxonomy of Experiments

- **Conventional Lab Experiment:** student pools, abstract framing, imposed set of rules
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Laboratory Experiments

Abstract task, Student populations, Imposed rules



- From 15 to 30 terminals + facility with separators
- Most used software: Ztree (.exe) and oTree (python + HTML)
- University Recruitment platform (e.g., ORSEE)
- Lab-Manager (do-it-all)

Lab Procedures

Before the experiment

- Ethical Approval
- Session planning + random sampling from platform (emails)

During the experiment

- Participants randomly seated (preserve anonymity)
- Receive instructions (written + read aloud by experimenter)
- No communication allowed

After the experiment

- Incentives (???)

Incentives

2 types in an experiment:

- To help recruit participants (show-up fee)
- To make decision making careful, hence, measure true preferences (performance based)

Big debate: Stated vs. Revealed Preferences

- Your decisions in an experiment have monetary consequences
- Properly designed incentives encourage participants to reveal true preferences.
- Difference w.r.t. vignette or survey experiments
- Yet, incentives do not always matter (Falk et al. 2024, Volinsky & Azar 2021 for a review)

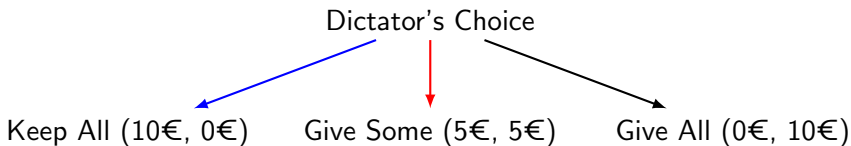
Some Interesting Examples of Lab Exp.

Prosociality

Exp. Econ is based on *game paradigms* : strategic situations in which we can use game theory to predict behavior.

Example: to measure individuals' generosity

- One could ask "How often do you help others?" (World Value Survey style)
- An experimentalist would say: **Dictator Game**



Prosociality & Environment Attanasi et al. 2025

Same paradigm, but different story behind: increase **voluntarily donations** to environmental organizations using AR visualizations



2 Conditions: Control (only Text), Treatment (Text + AR)

Cooperation in the Lab

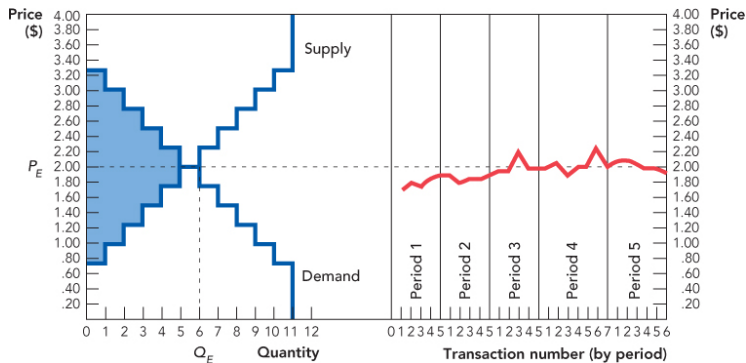
- **Test of game theory concepts:** Prisoner's Dilemma and Nash Equilibrium

	Cooperate	Defect
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

Table: Prisoner's Dilemma Payoff Matrix

Markets and Auctions

- Market Functioning: auctions



Natural Field Experiments



Natural Field Experiments



Figure: Reducing energy consumption using smart meters (Allcott 2011)

Natural environment, and subjects are not informed they are part of an experiment

- Heterogeneous populations (not students)
- Observing actual behavior in “natural” contexts
- Often requires partnerships and/or technology

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My view: Addressing different questions

- Lab is useful to understand *mechanisms*
- Field has highest validity
- One complements the other at different depths

Some Interesting Field Applications

Environment - Field Exp. - Guido et al. 2024

Objective: Reducing carbon footprint of internet data use

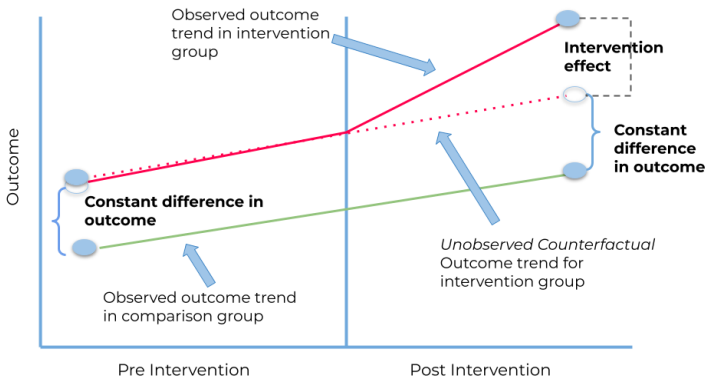
- 4/5G produces 4x more CO₂ than Wi-Fi

Experiment in a nutshell:

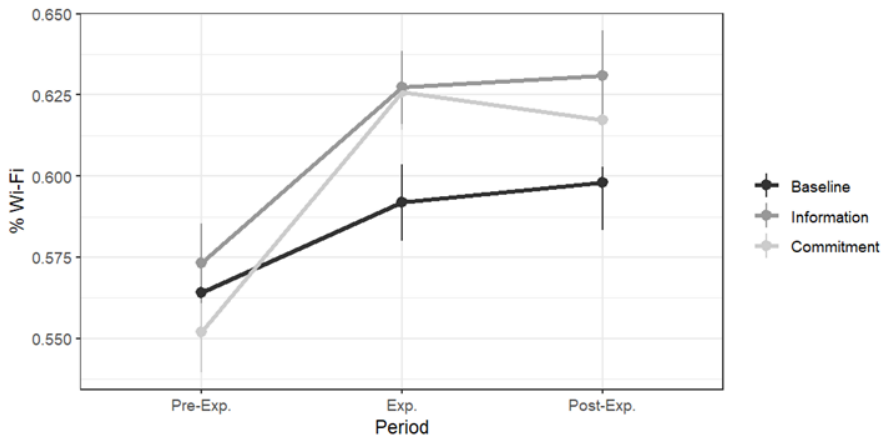
- Randomization across 3 conditions (between-subject)
- Each condition differ in how we "push" participants to reduce mobile consumption
 - Control (just observe)
 - Information (provide info on consumption consequences)
 - Commitment (set consumption goals)
- Use of a smartphone app to monitor data consumption
 - App retrieves data consumption **before** installation
 - so we have **Pre-** and **Post-**Intervention

Why I chose this example

Difference-in-Difference Design



Results



Other examples

Productivity - Gossnell et al. 2021: Testing managerial practices on Airlines Pilots' Behavior and Productivity

Electricity consumption - Allcott 2011: Testing whether social norms influence electricity consumption

An Experimentalist's checklist

An Experimentalist's checklist

- Ethical Approval
- Power analysis & Pre-Registration
- Open Science & Replicability

Power Analysis

Two main goals:

- Determine a-priori sample size, given an **effect size** (expected treatment effect)
 - Effect size based on past studies, pilots, or conventions (e.g., Cohen's d)
- Determine Minimum Detectable Effect (MDE), given a **sample size**
 - When constraints (e.g., budget) define our sample

Why it matters:

- Avoids underpowered studies (false negatives)
- Prevents waste from oversampling

Tip: Ex-post Power based on observed effects do not make sense (even when editors ask for it).

Pre-Registration

Goal: Improve transparency, credibility, and reproducibility.

What to include:

- Hypotheses and experimental design
- Sample size and power analysis
- Analysis plan (e.g., outcomes, regressions)

Platforms: OSF, AEA Registry, AsPredicted

Benefits:

- Limits p-hacking and HARKing
- Distinguishes confirmatory vs. exploratory analysis

Note: more journals demand it, even under the form of Registered Report

Open Science & Replicability

Main goal: Make your research accessible and replicable

Open science practices

- Make data & codes publicly available
- Make them understandable!
- Describe all details of your experiments

Replicability (cont.'d)

Open access, freely available online

Essay

Why Most Published Research Findings Are False

John P.A. Ioannidis

Figure: Ioannidis on the low rate of replicability in medical studies

- Replication crisis across disciplines
- Brian Nosek, researchers redid 100 studies in psychological science from three high-ranking psychology journals. Only 36 replicated.
- A study published in Science replicated 18 experimental studies published in Top5. 11 replicated. 70% reproducible.

Solution: **Pre-Registration + Open Science**

Failures and applied tips in the field

Main reasons leading to failing in the field

- Inappropriate research design
- Partner organization challenges
- Technical design flaws
- Low participation rate

From book by Karlan & Appel "*Failing in the Field: What We Can Learn When Field Research Goes Wrong*" + Personal Experience

1. Inappropriate research design

- Context matters: Research should be conducted in settings where the problem exists and the intervention is relevant.
- Timing issues: external events like politics or weather can disrupt research
- Feasibility: Ensure infrastructure and resources are available
- Knowing when to stop: do not be afraid to drop the study

Tip: Enlarge your toolbox

- **Qualitative methods:** Use interviews, focus groups, or field observations to understand the context and refine your research question.
 - **Combine and Iterate:** Combine methods and let early qualitative findings inform your quantitative design and vice-versa.
- **Descriptive Evidence:** No need for causality all the time! Stylized facts are precursor of bigger ideas

An example: Guido et al. 2024

Problem. Switching from mobile to Wi-Fi data consumption (offloading)

Before going to the field and design interventions, get familiar with the problem at hand!

Interviews with Providers and Users + Preliminary Survey: knowing whether people know about offloading and what interventions would be most effective

- more than 50% of subjects did not know about the offloading problem → Inform them!
- Mobile consumption correlates with impatience → Improve self control (commitments)

Co-design sessions: group-up approach to designing experiments

Did you know that mobile networks produce up to 4 times more CO₂ than Wi-Fi networks?

According to your declaration of **Mobile** data consumption, you will produce:

2290g of CO₂/per month

If you had used **Wi-Fi** only with the same amount of consumed data, you will produce only:

536g of CO₂/per month

By **switching** to **Wi-Fi** network, your **CO₂ emissions will be reduced** by a quantity equivalent to:

32h electricity heating	
219 Km in High speed train	
2928h of LED bulb on	

As a reminder, here is your breakdown of last week's data consumption:

38% via Wi-Fi
62% via Mobile

Example 2: Romaniuc, Guido et al. 2024

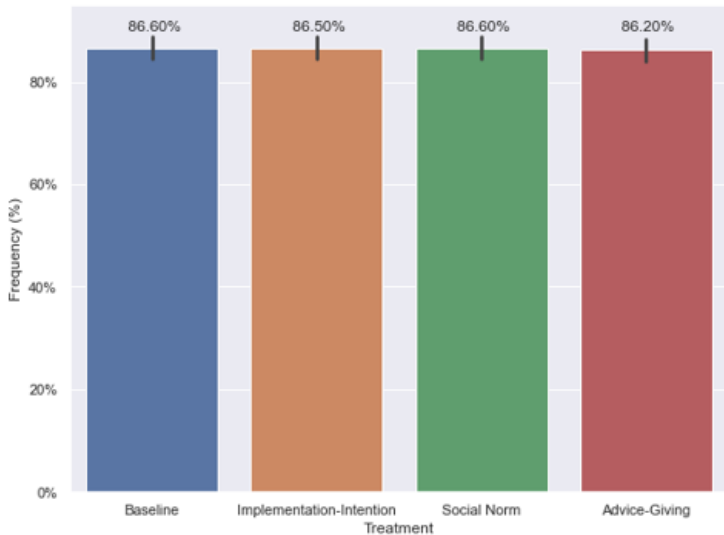
Problem. Nudging voter participation among young adults

Co-design sessions with experts



cont'd

Yet, no pre-experimental survey on baseline participation rates



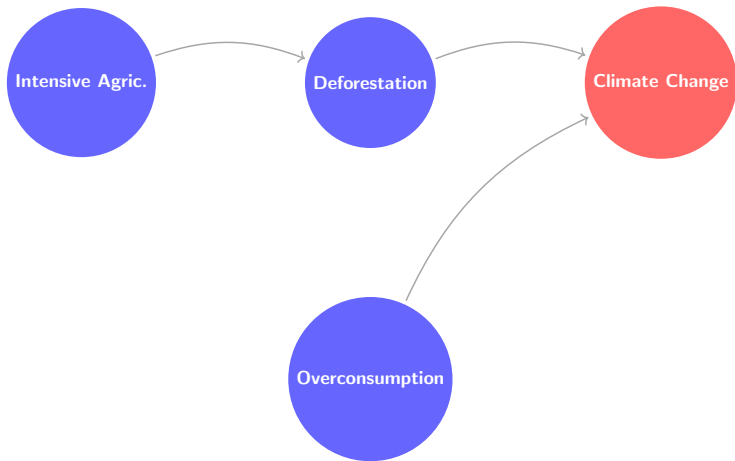
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An Example: Narratives of Climate Change

Ongoing work with Ph.D.s visiting PSB (Bea & Dany)

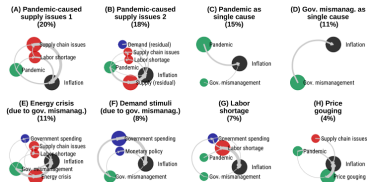
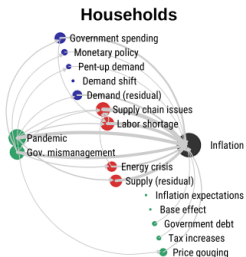
- **Factors causing climate change:** We use open-ended questions to elicit narratives on climate change
- Survey 10 countries (Europe, US, Asia) 3000+ respondents
- Transform narratives into graphs



"Overconsumption is causing climate change, as well as increasing demand in agriculture, which leads to deforestation"

An Example: Narratives of Climate Change

Find patterns, stylized facts and competing narratives



2. Partner organization

- It is hard to collaborate with organizations to implement programs and gather relevant data.
- Organizations are often afraid of experimenting (never say the “E” word)

My view:

- Show the advantages of using mixed methods
- Use experiments as a complementary tool to measure opportunity costs

3. Technical Design Flaws

- **Survey Design Errors:** Question wording, response scales and question order
- **Survey Length and Relevance:** Tendency to add unnecessary questions
- **Statistical Power and Sample Size**
- **Randomization Protocols**

4. Low participation rates

- Distinguish between *during intake* and *after treatment assignment*
- *During intake*: less participants than expected
 - sampling, hard-to-reach participants
 - wrong incentives
 - trust in the researchers or organizations involved
- *After treatment assignment*: potential bias in estimates
 - treatment makes harder to stay in the study?

Example: Loan Repayment in Nigeria (Guido & Romaniuc, in progress)

Trust towards researchers is important!

Objective. Measuring external validity of laboratory measures in the context of loan repayment

- Bank kindly provides customers' contacts
- Researchers (w/ local RA) send invitation emails
- Response rate: 5% !!!
- They thought we were scamming...
- Solutions: Involvement of local university + stricter protocol (phone calls with pre-defined script)

Future directions & Hot topics

3 Waves

- Wave 1 (90'): Testing (classic) theories

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- Wave 4: ?

Hot topics

Climate change, Green behavior and sustainability

- Reduce or change polluting behavior
- Correct misperceptions and wrong beliefs

Cognitive foundations in decision making

- The ways in which people attend to, remember, aggregate and trade off information
- Mental models and Narratives
- Motivated Reasoning

Digital Technologies

- Role of AI in decision making
- Digital Addiction (young adults)
- Environmental consequences of extensive use

Mixed methods: qualitative and quantitative

Sum Up!

- Beh. Sciences and Experiments: a fun way to get answers
- Mixed methods are the future and we can learn from each other
- I hope this talk will stimulate new research ideas and initiatives

Thank You

For the sake of open science, slides available here:
andreaguido.com